

Activation Energy and Catalysts

Answer key with worked solutions

A. Understanding

1. Definitions.

Activation energy: the minimum energy that colliding particles need in order to react.
Transition state: the highest-energy, unstable arrangement of atoms on the energy profile, between reactants and products.

2. Energy profile.

(a) ΔH is negative, so **exothermic**. (b) E_a (reverse) = $120 \text{ kJ/mol} + 40 \text{ kJ/mol} = \mathbf{160 \text{ kJ/mol}}$.
(c) The catalyst lowers both hills by 50 kJ/mol : E_a (reverse) = $70 \text{ kJ/mol} + 40 \text{ kJ/mol} = \mathbf{110 \text{ kJ/mol}}$; ΔH is **unchanged**, $\mathbf{-40 \text{ kJ/mol}}$.

3. Maxwell-Boltzmann.

Only molecules with energy $\geq E_a$ can react. At a higher temperature the distribution flattens and shifts to higher energy, so the **area beyond E_a grows much more than the average energy does**. A small temperature rise can therefore roughly double the number of molecules able to react (and they collide slightly more often too).

4. Catalysts.

A **homogeneous** catalyst is in the same phase as the reactants (e.g. H^+ ions catalysing ester hydrolysis in solution). A **heterogeneous** catalyst is in a different phase, usually a solid surface (e.g. iron in the Haber process, platinum in catalytic converters). A catalyst **does not change K**.

B. The Arrhenius equation

5. Factor from 300. K to 310. K.

$$\ln \frac{k_2}{k_1} = \frac{7.50 \times 10^4 \text{ J/mol}}{8.314 \text{ J/(mol}\cdot\text{K)}} \left(\frac{1}{300. \text{ K}} - \frac{1}{310. \text{ K}} \right) \\ = 0.970$$

$$\frac{k_2}{k_1} = e^{0.970} = \mathbf{2.64} \text{ (the units cancel inside the exponent)}$$

6. E_a from two rate constants.

$$\ln \frac{6.0 \times 10^{-4} \text{ s}^{-1}}{3.0 \times 10^{-5} \text{ s}^{-1}} = \ln 20. = 2.996; \frac{1}{500. \text{ K}} - \frac{1}{550. \text{ K}} = 1.818 \times 10^{-4} \text{ K}^{-1} \\ E_a = \frac{8.314 \text{ J/(mol}\cdot\text{K)} \times 2.996}{1.818 \times 10^{-4} \text{ K}^{-1}} \\ = 1.37 \times 10^5 \text{ J/mol} = 137 \text{ kJ/mol}$$

The rate constants have 2 significant figures, so report $E_a = \mathbf{1.4 \times 10^2 \text{ kJ/mol}}$ (137 kJ/mol before rounding).

7. k at 330. K.

$$\ln \frac{k_2}{k_1} = \frac{7.50 \times 10^4 \text{ J/mol}}{8.314 \text{ J/(mol}\cdot\text{K)}} \left(\frac{1}{300. \text{ K}} - \frac{1}{330. \text{ K}} \right) \\ = 2.734$$

$$k_2 = 2.0 \times 10^{-4} \text{ s}^{-1} \times e^{2.734} = 3.1 \times 10^{-3} \text{ s}^{-1} \text{ (} 3.08 \times 10^{-3} \text{ s}^{-1} \text{ before rounding to 2 significant figures)}$$

C. Challenge

8. Effect of the catalyst.

With A unchanged, $\frac{k_{\text{cat}}}{k} = e^{(E_a - E_{a,\text{cat}})/RT}$:

$$\frac{(7.50 - 5.00) \times 10^4 \text{ J/mol}}{8.314 \text{ J/(mol}\cdot\text{K)} \times 298 \text{ K}} = 10.09$$

$$\frac{k_{\text{cat}}}{k} = e^{10.09} = 2.4 \times 10^4 \text{: about 24 000 times faster.}$$

9. Arrhenius plot.

Slope = $-\frac{E_a}{R}$, so $E_a = 6.65 \times 10^3 \text{ K} \times 8.314 \text{ J}/(\text{mol}\cdot\text{K}) = 5.53 \times 10^4 \text{ J/mol} = \mathbf{55.3 \text{ kJ/mol}}$
(the units of K cancel).